

ANKUR MISHRA

Machine Learning Engineer | Data Science & Computer Vision

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PROFESSIONAL SUMMARY

Aspiring Machine Learning and Data Science professional with hands-on experience in Computer Vision, Deep Learning, Data Analytics, and Research Engineering. Skilled in building end-to-end ML pipelines, conducting exploratory data analysis on real-world datasets, and developing AI solutions for agriculture, biological image analysis, and business analytics. Co-authored published research on plant species identification using transfer learning CNNs. Proficient in Python, TensorFlow, Scikit-learn, OpenCV, and LaTeX. Passionate about bridging academic research and industry applications through practical, data-driven problem solving.

TECHNICAL SKILLS

Languages & Libraries: Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, OpenCV, Scikit-Image

ML / DL Techniques: CNN, Transfer Learning (VGG16, VGG19, ResNet50, InceptionV3, EfficientNet, MobileNetV2), Feature Engineering, Model Evaluation, EDA, NLP (CountVectorizer, Naive Bayes), ROC-AUC Analysis

Databases & Tools: SQL, SQLite, Database Management, Jupyter Notebook, LaTeX, Git

Core Competencies: Research Methodology, Technical Writing, Literature Review, Data Preprocessing, Predictive Modeling, Computer Vision, Medical & Biological Imaging

EXPERIENCE

Research Assistant Intern

Feb 2026 – Aug 2026

Knowledge Square Services LLP · Remote / Vadodara

- Developed and implemented ML and data analytics pipelines for research and industry-oriented projects.
- Conducted EDA, data preprocessing, feature engineering, and model evaluation on real-world datasets.
- Contributed to research paper writing, technical documentation, editorial review, and LaTeX manuscript preparation.
- Translated complex technical concepts into client-friendly presentations, tutorials, and recorded demonstrations.
- Collaborated on AI, computer vision, and data science projects with a focus on practical research methodologies.
- Generated analytical reports and insights to support data-driven decision-making.

Junior Research Fellow

2024 – Present

Parul University · Vadodara, Gujarat

- Designed and evaluated CNN-based pipelines for image preprocessing and HLB (Huanglongbing) plant disease classification.
- Benchmarked six deep learning architectures (VGG16, VGG19, ResNet50, InceptionV3, EfficientNet, MobileNetV2) using Accuracy, F1-score, and ROC-AUC.
- Contributed to a peer-reviewed research paper on optimising deep learning models through structured feature extraction.
- Coordinated with senior researchers to design methodologies and ensure adherence to project objectives.
- Engaged research participants and facilitated high-quality data collection for experimental studies.

Data Analytics Job Simulation

January 2026

Tata iQ · GenAI-Powered Data Analytics · Forage · Credential: AnLdtkT5E6xNyy6Ez

- Conducted EDA using GenAI tools to assess data quality, identify risk indicators, and structure insights for predictive modeling.
- Proposed a no-code predictive modeling framework to assess customer delinquency risk using GenAI-structured evaluation criteria.
- Designed an AI-driven collections strategy incorporating agentic AI, ethical AI principles, and regulatory compliance frameworks.

PROJECTS

Plant Species Identification using Leaf Venation Patterns CNN · Transfer Learning · ResNet50 · MobileNetV2 · EfficientNetB0

- Co-authored research on automated plant species identification using leaf venation features and transfer learning across 15 species (1,125 images).
- Achieved best classification accuracy of 94.67% using EfficientNetB0 — highest among all evaluated architectures.
- Developed and compared deep learning models; contributed to a peer-reviewed publication.

Leaf Venation Extraction and Preprocessing Pipeline OpenCV · Scikit-Image · Python · Computer Vision

- Designed an end-to-end image preprocessing pipeline for venation enhancement, segmentation, contour extraction, and skeletonization.
- Generated structured inputs for CNN and Graph Neural Network (GNN) classification workflows.
- Improved feature visibility and dataset quality for biological image analysis applications.

Plant Salinity Stress Analysis using Deep Learning Python · VGG19 · InceptionV3 · ROC-AUC Analysis

- Analyzed real-world spectral plant data to investigate salinity stress using deep learning techniques.
- Evaluated VGG19 and InceptionV3 on multiple spectral representations including Green-IR combinations and spectral band ratios.
- Assessed model performance using ROC-AUC metrics and quantified the impact of spectral preprocessing on classification outcomes.

Customer Marketing Campaign Analysis Python · Pandas · NumPy · Matplotlib · Seaborn · Automated EDA

- Performed comprehensive analysis of customer demographics, purchasing behaviour, campaign engagement, and spending trends.
- Identified customer segments, data quality issues, and campaign response trends through structured exploratory data analysis.
- Developed automated EDA reporting workflows to accelerate business insight generation and reduce analyst turnaround time.

Spam Email Classification with Automated EDA Python · Scikit-learn · CountVectorizer · Multinomial Naive Bayes

- Built a text classification pipeline for spam detection using NLP vectorization and Multinomial Naive Bayes classification.
- Integrated automated EDA reporting for dataset profiling, class distribution analysis, and feature exploration.

EDUCATION

Bachelor of Technology (B.Tech) — Artificial Intelligence & Data Science	2024 – 2028 (Expected)
Parul University, Vadodara, Gujarat	GPA: 7.50 / 10.0 (Current)

CERTIFICATIONS

Smart Tools in Artificial Intelligence

Parul University

What is Data Science?

IBM · Coursera

Search Engine Optimization (SEO) with Squarespace

Coursera

GenAI-Powered Data Analytics Job Simulation

Tata iQ · Forage

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